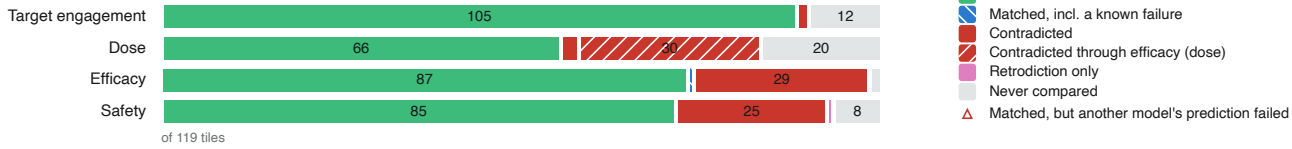
a



Panel b: the top 15 diseases by global DALYs in 2021 (WHO Global Health Estimates), each with the mechanism that had the most programs.

b

		Target engagement	Dose	Efficacy	Safety
#1	**type 2 diabetes mellitus** GLP1R agonists (12)	**2011** Cynomolgus monkeys and rat islets (dulaglutide)	**2019** Beagle dog oral bioavailability of semaglutide/SNAC tablets	**2022** GLP1R Gly168Ser genotype, GLP-1 infusion clamp in healthy volunteers	**2016** GLP1R Ala316Thr carriers (human genetics)
#2	**COPD** CHRM3 antagonists (6)	**2018** Human isolated bronchi: carbachol-evoked contraction, potency and washout offset	**2018** Human isolated bronchi: carbachol-evoked contraction, potency and washout offset	**2018** Human isolated bronchi: carbachol-evoked contraction, potency and washout offset	**2016** Rat bronchoprotection and antisialagogue; dog bronchospasm △ Stopped; preclinical unexplained
#3	**HIV** HIV RT (NNRTI) inhibitors (3)	**2012** HIV-1 culture vs wild-type and NNRTI-resistant virus, resistance selection, protein-adjusted IC90	**2020** In-silico doravirine PK model with rifabutin from the single-dose DDI, setting 100 mg twice daily	**2012** HIV-1 replication in cell culture (MT-4/PBMC) against wild-type and NNRTI-resistant virus	**2012** Rodent and monkey tox as filed for fosdevirine and doravirine Seizures no tox species predicted
#4	**NSCLC** EGFR inhibitors (18)	**2017** T790M/sensitising-mutant NSCLC cells and mice	**2019** PC9 brain-metastasis mice; monkey osimertinib PET	**2023** Patient-derived lung cancer organoids △ EXCLAIM-2 fails primary PFS	**2015** hERG, rat and dog tox, dog telemetry QT prolongation the package cleared
#5	**migraine disorder** CGRP receptor antagonists (4)	**2014** Rhesus capsaicin-induced dermal vasodilation (CIDV)	**2014** Rhesus CIDV PK/PD target Predicted 20 mg; needed 200 mg	**2013** Primate CGRP-induced facial blood flow (rimegepant / BMS-927711)	**2026** Mouse MCAO stroke, gepant pretreatment △ Animal tox missed MK-3207 liver
#6	**Alzheimer disease** BACE1 inhibitors (4)	**2015** PDAPP mice and cannulated beagle dogs (brain/CSF Aβ)	**2019** Dog CSF Aβ PK/PD scaled to humans Target reached, no benefit	**2019** A673T variant carriers Genetics said protect; drug worsened	**2019** Mice chronically dosed with BACE1 inhibitors △ Liver injury missed by tox
#7	**asthma** PTGDR2 antagonists (9)	**2016** Human eosinophils and Th2 cells in vitro	**2019** Eosinophil DP2 potency Target reached; exacerbations unchanged	**2013** Allergen challenge Challenge said yes; FEV1 trials missed	**2012** Off-target selectivity panels; animal toxicology unpublished
#8	**breast cancer** CDK4/6 inhibitors (6)	**2016** Human tumour xenografts in mice (phospho-Rb Ser780)	**2017** Abemaciclib xenograft PK/PD (pRb to growth) △ Brain exposure OK, no response	**2014** Human breast cancer cell-line panel (47 lines) △ Brain-metastasis trial missed	**2017** Dog cardiovascular telemetry (QTc) with hERG assays △ Missed liver injury and clots
#9	**osteoarthritis** MAPK14 inhibitors (3)	**2012** Rat iodoacetate joint degeneration + carrageenan thermal hyperalgesia (SB-203580, VX-745)	**2009** Rat and cynomolgus endotoxin-induced TNF-alpha PD model (PH-797804)	**2012** Rodent rheumatoid-arthritis models, ex-vivo LPS whole-blood cytokine PD (ARRY-797)	**2014** Rat repeat-dose intra-articular toxicology (FX005)
#10	**colorectal cancer** KDR (VEGFR) inhibitors (3)	**2014** Mouse lung VEGF-induced pVEGFR2 (fruquintinib)	**2014** Mouse lung VEGF-induced pVEGFR2 (fruquintinib)	**2017** Human colorectal patient-derived xenografts (PDX) in mice	**2023** Dog CV safety, rat/dog tox Dogs silent; 14% grade 3-4 hypertension
#11	**chronic hepatitis B virus infection** HBV RNA antisense / siRNA (3)	**2022** Chronically HBV-infected chimpanzees	**2018** HBV-infected human liver-chimeric mice (PXB uPA/SCID with human hepatocytes)	**2023** HBV-genome mice (pHBV transient/transgenic, later AAV-HBV)	**2023** HBV-infected human liver-chimeric mice (PXB uPA/SCID with human hepatocytes)
#12	**hepatocellular carcinoma** VEGFR multikinase inhibitors (4)	**2015** Human HCC cell-line and patient-derived xenografts in immunodeficient mice	**2018** Lenvatinib population PK and exposure-response model in HCC, choosing weight-based starting dose	**2020** Mouse HCC with diverse transposon-delivered genetic drivers, lenvatinib vs sorafenib susceptibility	**2015** GLP repeat-dose toxicology in rat plus dog or monkey
#13	**schizophrenia** PDE10A inhibitors (5)	**2019** Rat and monkey ketamine-challenge phMRI/EEG reversal	**2019** Rodent occupancy dosing Predicted-adequate exposure, no effect	**2017** Ketamine challenge Challenge passed; phase 2 PANSS missed	**2017** Rat catalepsy, prolactin and glucose vs D2 antagonists EPS in patients at clinical dose
#14	**gastric cancer** PARP1/2 inhibitors (3)	**2023** Gastroesophageal PDX, BRCA2-mutant, PARP inhibitors	**2023** Gastroesophageal PDX, BRCA2-mutant, PARP inhibitors	**2026** Gastroesophageal PDX, BRCA2-mutant, PARP inhibitors △ Hypoxia-sensitisation bar missed	**2017** Rat and dog general toxicology, up to 26 weeks (olaparib)
#15	**hepatitis C** HCV NS3/4A protease inhibitors (13)	**2013** Chronically HCV-infected chimpanzee (genotype 1a/1b)	**2013** Replicon EC99 vs rat plasma and liver exposure	**2016** Human hepatocyte chimeric mouse (uPA/SCID) infected with patient HCV genotype 1b	**2016** Rodent and dog repeat-dose toxicology